

INSIGHTWELLNESS AI

Milestone 3 — Final Presentation

INFO9023 Machine Learning Systems Design · ULiège · Spring
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Use case & Dataset

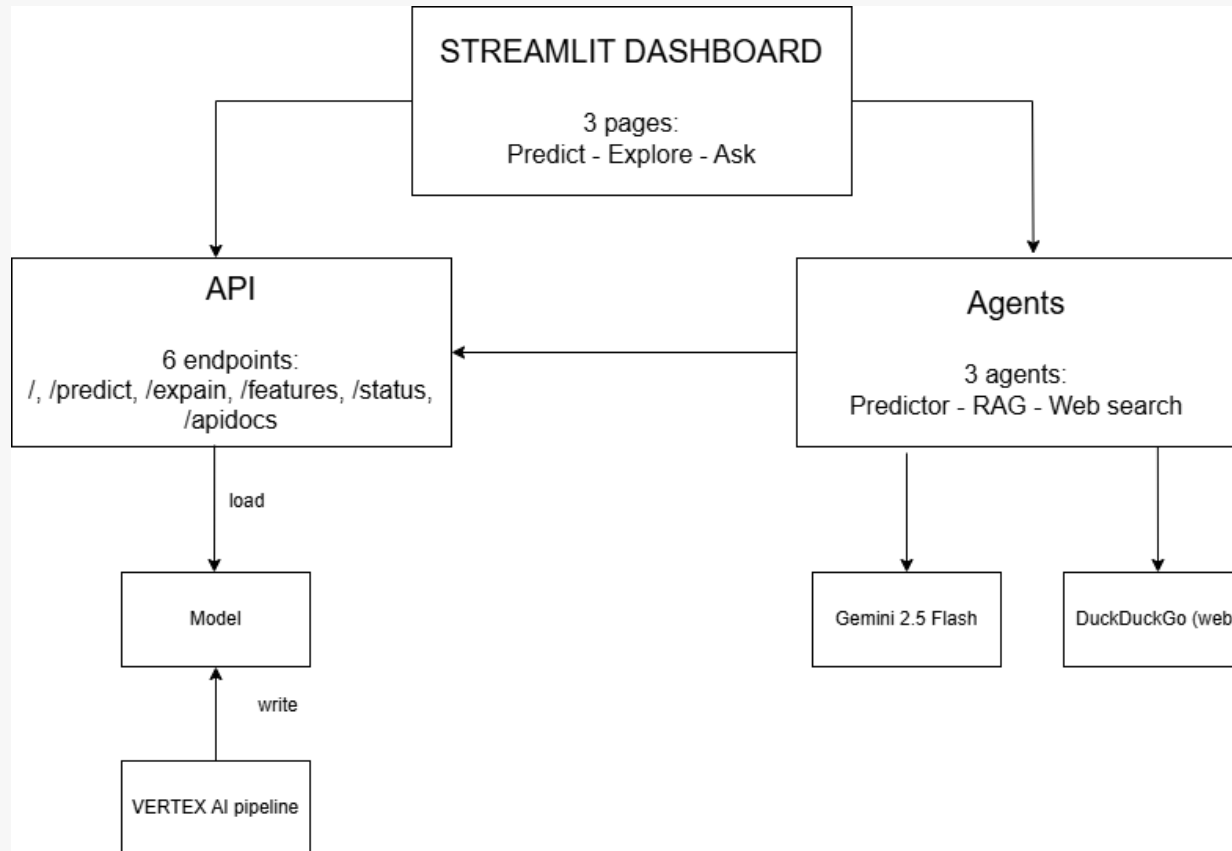
- **Early obesity risk detection from lifestyle data**
- 16 features (diet, activity, daily habits)+ 1 target · 7 classes
- 77% SMOTE-synthetic, 23% original survey

Data leakage decision

- Height/Weight model scored 96% accuracy
- Label is BMI-derived from height & weight
- Using them = label leakage, not skill
- Final model (HistGradientBostingClassifier) uses lifestyle features only

Metric	Current Model	Height/Weight Model
Accuracy	82.03%	96.45%
F1-macro	81.80%	96.33%
CV F1 mean	81.71%	96.18%
CV F1 std	±2.31%	±0.93%

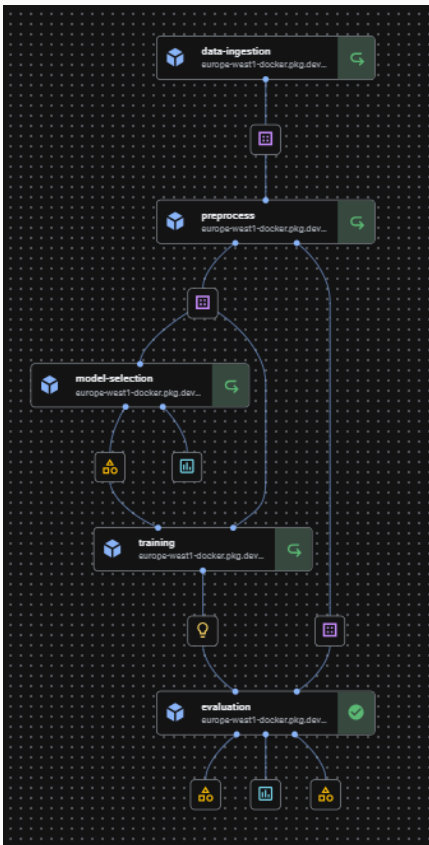
System architecture



API

- **6 endpoints:**
 - /
 - /status
 - /features
 - /predict
 - /explain
 - /apidocs
- **Cloud Run · scale-to-zero · max 50 instances**

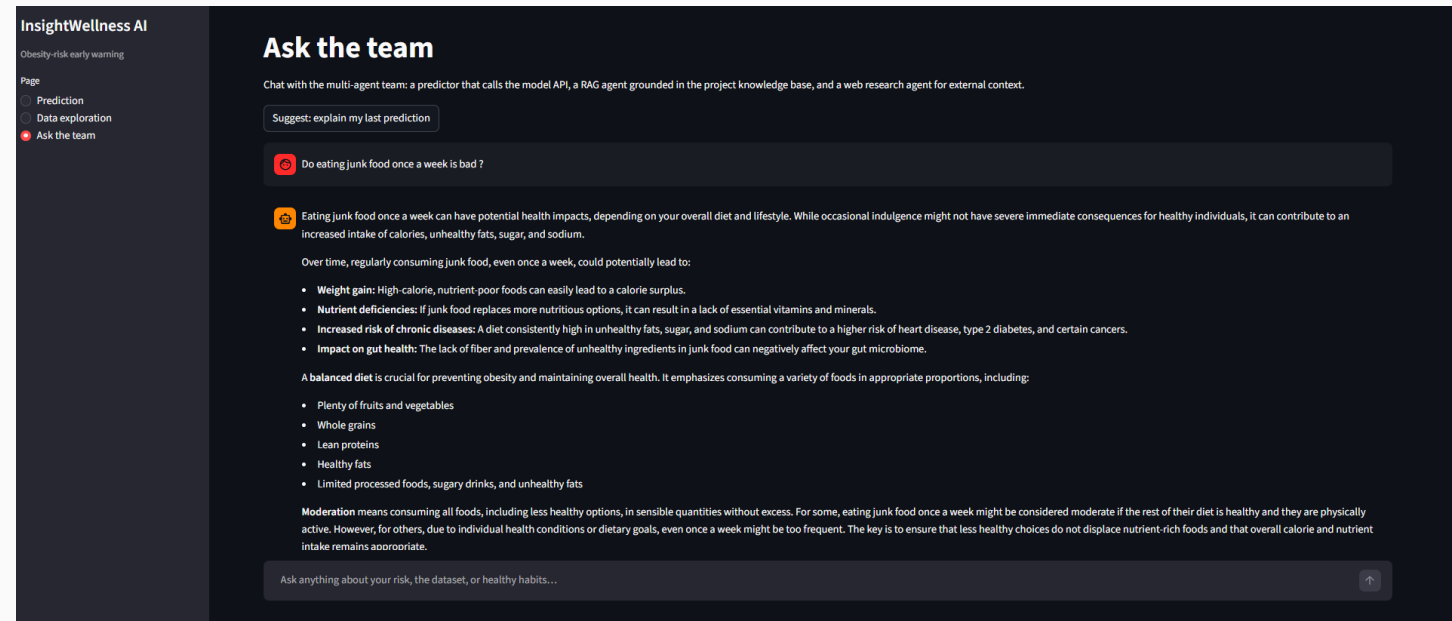
Vertex AI Pipeline



- **Data ingestion:** Collect data from BigQuery
- **Preprocess:** Cleaning, feature engineering and train/test split.
- **Model selection:** Hyperparameter tuning.
- **Training:** Train on train data.
- **Evaluation:** Produces performance metrics using test data.

Streamlit dashboard

- Prediction form → /predict and (or) /explain
- Feature distributions with user overlay
- Ask the team (multi-agent chat)



Multi-agent layer

- Agno Team coordinates three agents
- Predictor agent: calls /predict + /explain
- RAG agent on lifestyle/nutrition KB
- Web search agent (DuckDuckGo)
- Backed by Gemini

CI/CD Pipeline

- **Ci.yml:** Code quality, pre-commit hooks and pytest
- **Vertex_pipeline.yml:** Build Docker image, push to artifact registry and trigger to vertex ai pipeline.
- **Streamlit.yml:** Build Docker image, push to artifact registry and deploy to cloud run.
- **Api.yml:** Build Docker image, push to artifact registry and deploy to cloud run.
- **Agent_api.yml:** Build Docker image, push to artifact registry and deploy to cloud run.

DEMO

